

Centripetal Motion & Gravitation Part 2

Roller Coasters, Normal Force, and Orbital Period

Name: _____

Date: _____

Constants (use unless told otherwise):

$g = 9.8 \text{ m/s}^2$, $G = 6.67 \times 10^{-11} \text{ N} \cdot \text{m}^2/\text{kg}^2$, $M_E = 5.97 \times 10^{24} \text{ kg}$, $R_E = 6.37 \times 10^6 \text{ m}$

Brief Recap / Mini-Lesson

1) What is Uniform Circular Motion (Centripetal Motion)

- Is velocity constant?
- How can we have acceleration?
- What are my dependent variables?
- How does my centripetal acceleration formula look like?
- How does my centripetal force formula look like?
- Is centripetal force its own force?
- What are some examples of centripetal forces?
- How would my centripetal acceleration and centripetal force look like graphically?

2) Universal Law of Gravitation

- What are my three pillars/laws of the universal law of gravitation?
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- Based on these laws, how can we construct a formula to model this?
- How can we relate the acceleration due to gravity on Earth with respect to this formula?

3) Centripetal Motion: Roller Coasters

Suppose we have an object at rest. What are the forces acting on the object?

Now suppose an object moves in a **vertical circle**, (a loop in a roller coaster). How would my forces look like now?

How would my forces look like at the bottom of the roller coaster?

How would my forces look like at the top of the roller coaster?

How would my force summations in the y-direction look like for the first scenario?

How would my force summations in the y-direction look like for the second scenario?

Does a person feel (heavier/lighter) in the first scenario?

4) Orbital Speed and Orbital Period

If

$$F_g = F_c$$

How can I find the orbital speed?

How can I find the time it takes to complete one orbit?

Guided Example: Roller Coaster Normal Force (Bottom vs Top)

A rider of mass 60 kg goes over a circular hill (top) and then through a circular dip (bottom). Both have radius $r = 20$ m. At the top of the hill the speed is 12 m/s, and at the bottom of the dip the speed is 18 m/s.

E1. Find the normal force on the rider at the top of the hill.

E2. Find the normal force on the rider at the bottom of the dip.

E3. Which normal force should be larger, top or bottom? Explain in one sentence.

Practice Problems

Directions: Show all work and include units.

1. A 0.40 kg rubber stopper moves in a horizontal circle of radius 0.80 m at constant speed 6.0 m/s.
 - (a) Find a_c .

 - (b) Find F_c .

2. A 1500 kg car rounds a flat circular turn of radius 55 m. The coefficient of static friction is $\mu_s = 0.70$.
 - (a) What is the maximum speed without slipping?

 - (b) What provides the centripetal force?

3. A rider of mass 72 kg travels through the **bottom** of a dip that is part of a circle of radius 25 m. At the bottom, the speed is 16 m/s.
 - (a) Draw a free-body diagram (FBD).

 - (b) Find the normal force N at the bottom.

4. The same rider from Problem 3 goes over the **top** of a circular hill of radius 25 m with speed 10 m/s.
 - (a) Draw a free-body diagram (FBD).

(b) Find the normal force N at the top.

(c) Will the rider feel “lighter” at the top or bottom? Explain using your normal force results.

5. Two objects with masses $m_1 = 8.0\text{ kg}$ and $m_2 = 3.0\text{ kg}$ are separated by 0.50 m .

(a) Find the gravitational force between them.

(b) If the distance doubles, by what factor does the force change?

6. A satellite orbits Earth in a **circular orbit** at altitude 400 km above Earth’s surface.

(a) What is the orbital radius r from Earth’s center.

(b) What is the orbital speed?

(c) Find the orbital period (time)?